

Charan Sai Ponnada

+91-8639269120 | charansaiponnada06@gmail.com | linkedin.com/in/charansaiponnada | github.com/charansaiponnada | charansaiponnada.vercel.app

AI Engineer & SDE · LLM Pipelines · RAG Systems · Full-Stack SaaS · IEEE Author · PPO → Full-Time · IIT Hyderabad Finalist (7,600+)

EXPERIENCE

Aynstyn Technologies Pvt. Ltd. (aynstyn.com)

Remote

AI Engineer Intern → Software Engineer (Full-Time, PPO)

Jan 2026 – Present

- Improved LLM response reliability across **3+ production workflows** by conducting systematic black-box testing — surfaced critical edge cases and stabilized AI outputs across live platform features
- Independently designed and built **LabSoft** (Next.js, PostgreSQL) from zero as sole developer — shipped **2 production modules** (Patient Management & Reporting) end-to-end with zero external support
- Architected **Aynstyn Intel** to provide admins real-time learner visibility — tracks Bloom's taxonomy distribution, knowledge-gap heatmaps, and funnel analytics across course cohorts
- Driving end-to-end development of the **Aynstyn mobile app** (React Native) — performed **LLM prompt tuning** to reduce inconsistent AI outputs; platform advanced to final testing phase

PROJECTS

VIVIRITY Intelli-Credit | Credit Risk Intelligence System

github.com/charansaiponnada/VIVIRITY

Python, LangChain, RAG, OpenAI API, Pandas, NumPy, FastAPI

- Built end-to-end AI credit intelligence system — **2nd Place** among Top 10 Finalists at IIT Hyderabad AI/ML Hackathon (YUVAAN 2026) from **7,600+ registrations**
- Automated **500+ page** annual report ingestion to complete in under **5 minutes** — reduced financial analysis turnaround from days to minutes
- Engineered **non-embedding RAG** pipeline achieving **98% lower API overhead** vs. vector-based retrieval systems
- Extracted **50+ financial risk indicators** via multi-agent decision pipeline for end-to-end loan risk assessment

AyurMind | Domain-Specific RAG System

github.com/charansaiponnada/AyurMind

Python, FAISS, LangChain, Sentence Transformers, FastAPI, HuggingFace

- Built RAG system over **1,200+ structured** Ayurvedic knowledge entries for accurate domain-specific question answering
- Achieved **84% Top-3 retrieval accuracy** via multi-stage optimized retrieval pipeline
- Reduced hallucinated responses by **45%** through multi-layer validation and output filtering mechanisms
- Improved expert-rated answer quality from **3.1 to 4.4/5** through iterative pipeline refinement and re-ranking

RESEARCH

Vision-Language Based Real-Time Assistive Navigation System | Primary Author

ISAECT 2025

Python, PyTorch, BLIP, LoRA, HuggingFace, Raspberry Pi, Edge-Cloud

- Built real-time BLIP-based assistive navigation system for visually impaired users in complex Indian urban environments — deployed on Raspberry Pi with edge-cloud hybrid architecture
- Fine-tuned BLIP via **3-stage LoRA training** on a custom **427-scene** Indian traffic dataset — achieved **+15.6%** improvement across BLEU, METEOR, ROUGE, and semantic similarity
- Designed full pipeline: dataset curation → model training → evaluation framework → edge deployment on Raspberry Pi
- Published at IEEE ISAECT 2025, Mohali — model on HuggingFace; code open-sourced on GitHub | DOI: **10.1109/ISAECT68904.2025.11318802**

TECHNICAL SKILLS

Programming: Python, Java, C, SQL, JavaScript

AI/ML: PyTorch, TensorFlow, Keras, SciPy, Model Fine-Tuning (LoRA), Evaluation Metrics

LLM Systems: RAG, Non-Embedding Retrieval, Prompt Engineering, LLM Evaluation, LangChain, FAISS

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, Microservices

Frontend: Next.js, React, React Native

Data Engineering: Pandas, NumPy, MySQL, MongoDB, PostgreSQL

Cloud & Deployment: AWS, GCP, Model Serving, Edge Deployment, Scalable Inference

Tools: Git, Docker, HuggingFace, Sentence Transformers

EDUCATION

Velagapudi Ramakrishna Siddhartha Engineering College

B.Tech in Artificial Intelligence and Data Science — CGPA: 8.76/10

Vijayawada, India

Sept 2023 – May 2027